July 7, 2023

Reflection 2 : Due tonight by midnight

Conceptual HW 2 : Due Sunday by midnight

Lab : Tuesday

Written HW 2 : Due Tuesday by midnight

Office Hours

Mondays Noon-2pm
Thursdays 2-3pm

	Monday	Tuesday	Wednesday	Thursday	Friday
Week 2	July 3 No Class	July 4 No Classes!	July 5 V2 CH 5 HW	July 6 V2 CH 6 Studio : Fie alcs	July 7 V2 CH 6 Reflection
Week 3	July 10 V2 CH 6	July 11 Homework 2 Lab W3 - Using E	July 12 V2 CH 7	July 13 V2 CH 7 Studio : Gauss's Law	July 14 V2 CH 7 Reflection
Week 4	July 17 V2 CH 8	July 18 Homework 3 Lab W4 - Electric Potential	July 19 V2 CH 8	July 20 V2 CH 9 Studio : V and C	July 21 Reflection Quiz 1&2: CH 5-7

Chapter 6

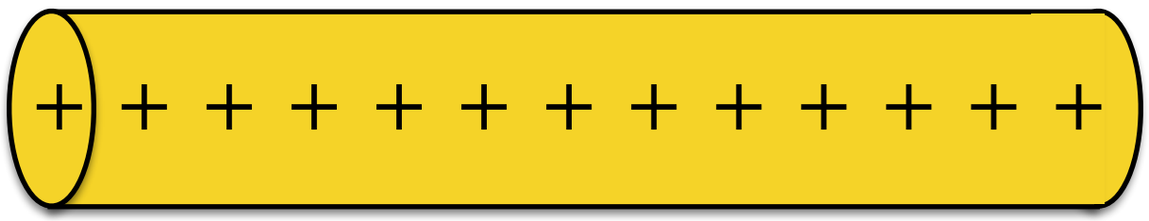
Gauss's Law

Symmetry

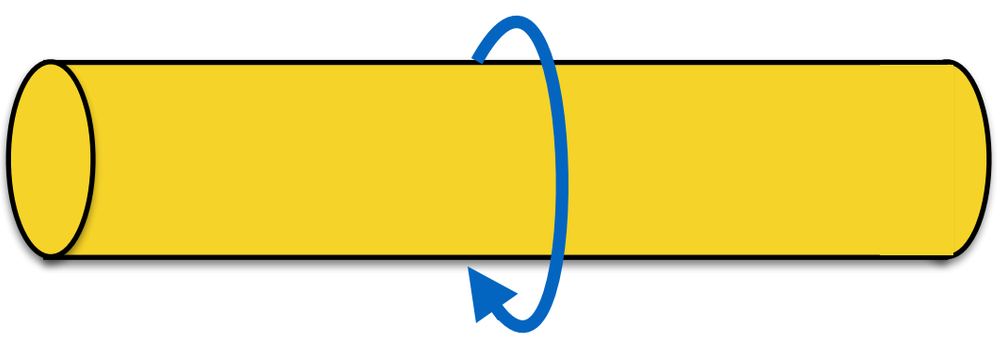
§ 6.3
Lecture 07

Symmetry

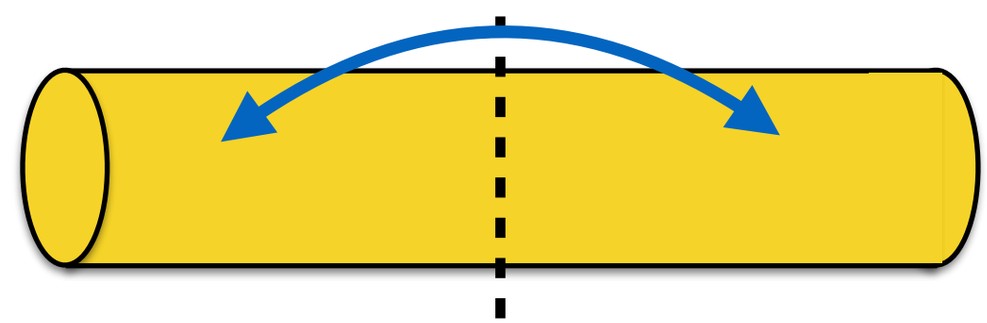
Changes made to a system that result in the system <u>appearing identical</u> to how it was previously.



Types of symmetry :



Rotational

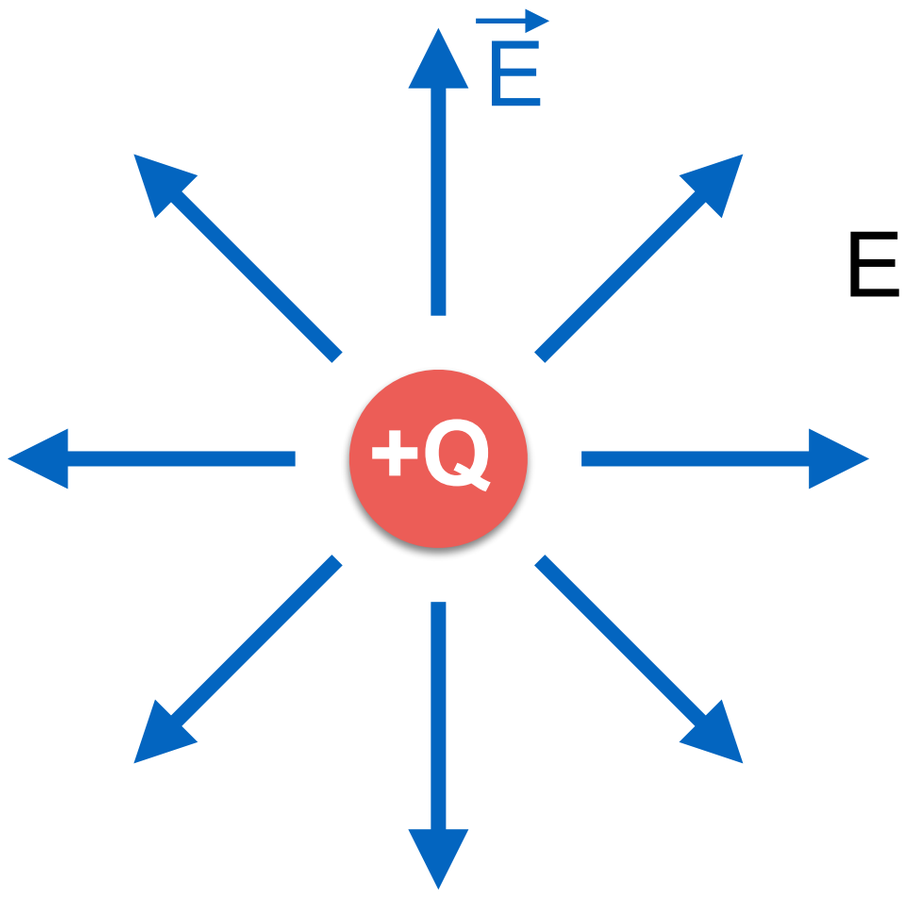


Reflection

Translational

Using Symmetry

Symmetry tells you what the field could look like.



E expected to: Be radially outward

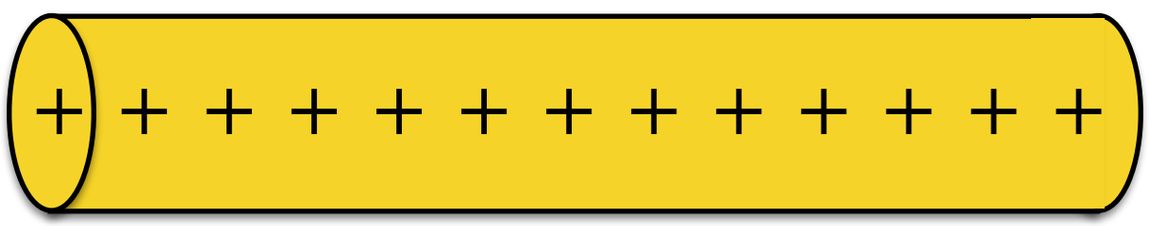
Have same magnitude at equal distances from the sphere

The field should have the <u>same</u> symmetries as the charges.

Using Symmetry

Charged Rod

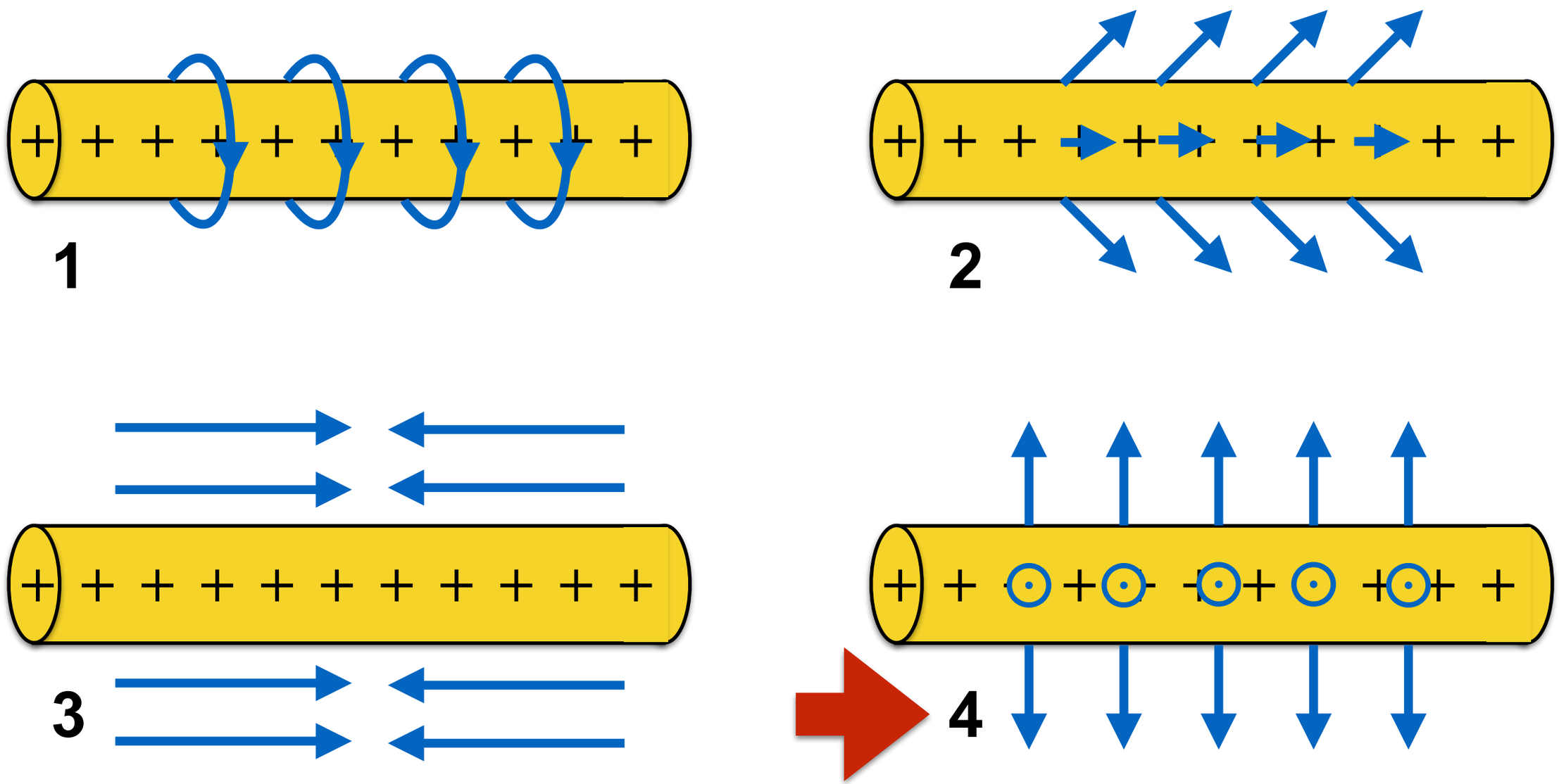
What Gaussian surface is most useful for calculating the field created by a very long (infinite), positively charged rod?



Using Symmetry

Charged Rod : Field Lines

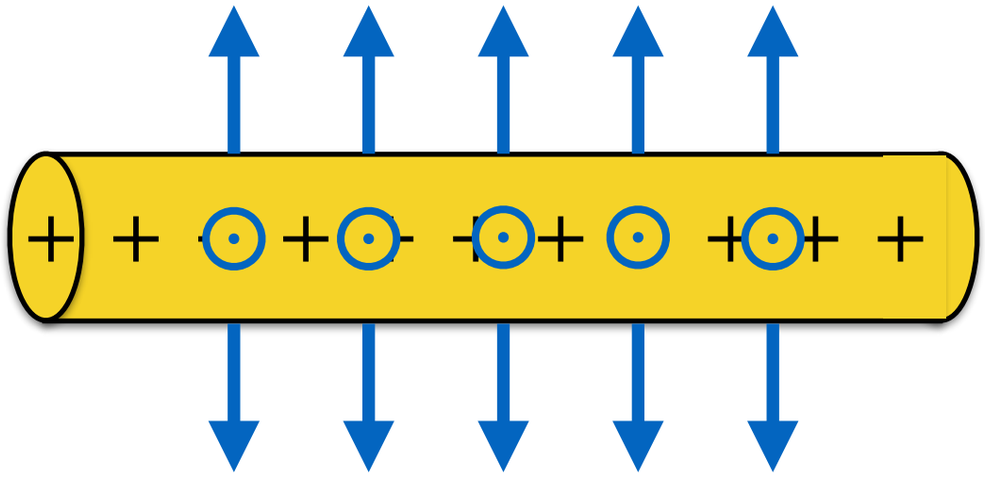
Using symmetry and what you recall about field lines, which could be the shape of the *electric field* for the long, charged rod?



Using Symmetry

Charged Rod : Variables

On what variables might the electric field depend outside of the (infinite) rod?

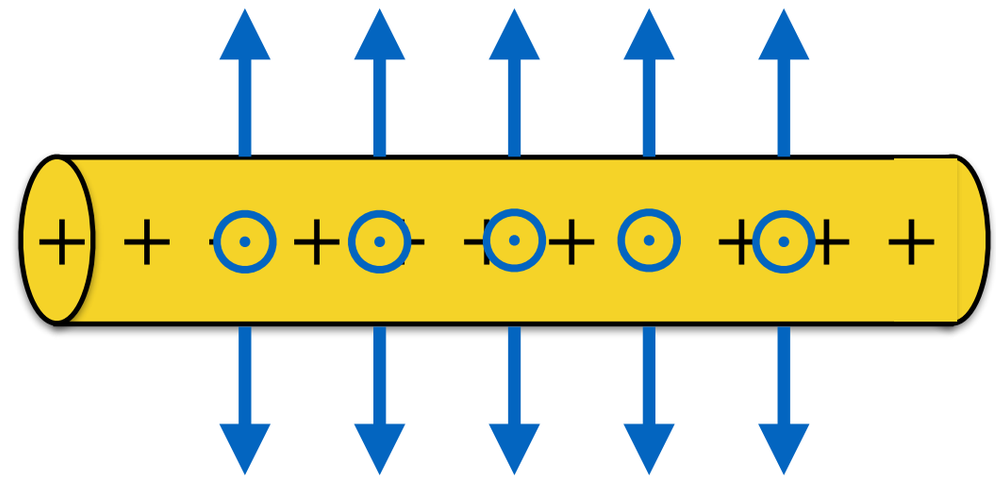


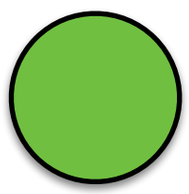
1. Distance from (center of) the rod
2. Radius of the rod
3. Distance along length of the rod
4. All of the above
5. 1 and 2
0. Don’t wait on my response

Using Symmetry

Charged Rod : Gaussian Surface

What shape for a gaussian surface might you want to pick for the rod of charge so that **E·dA** is constant?

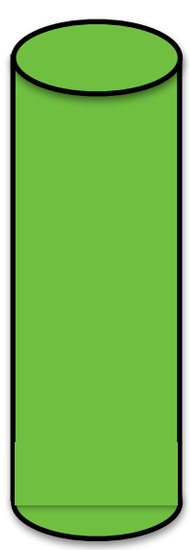


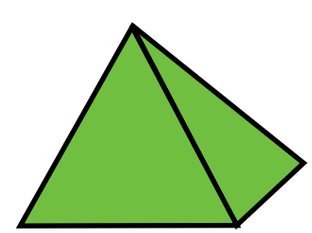


1. Sphere

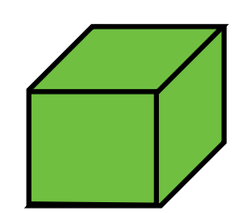
3. Horizontal Cylinder

2. Vertical Cylinder





4. Pyramid



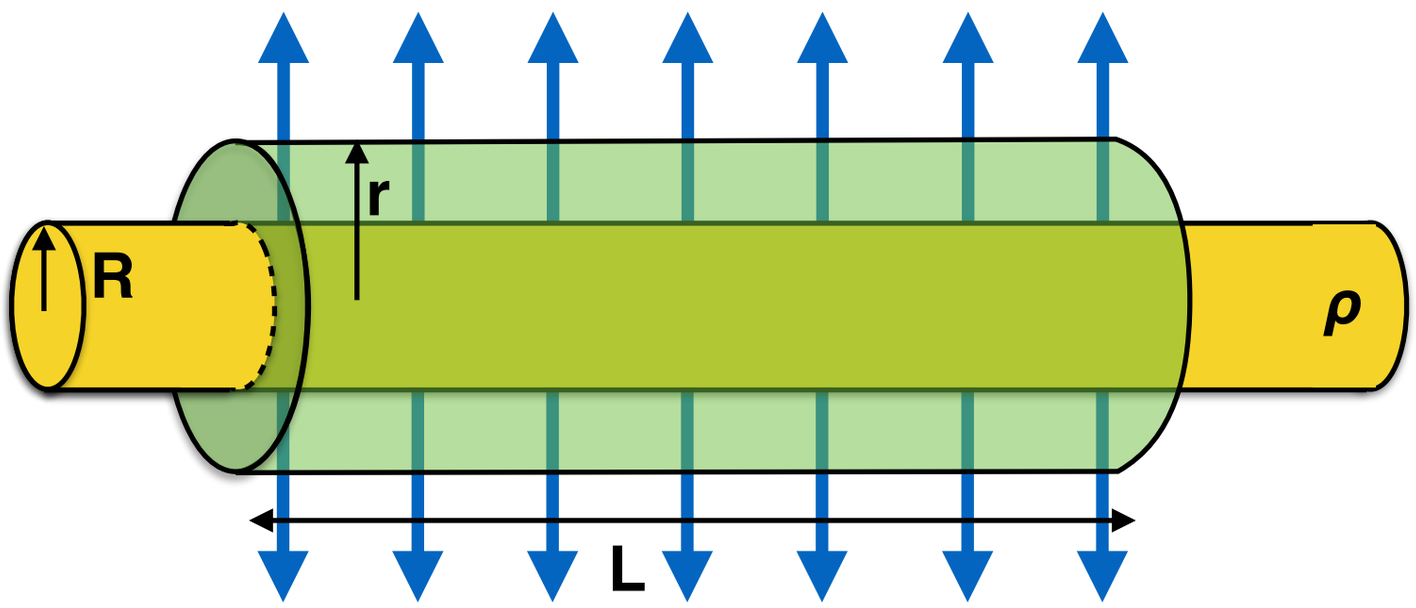
5. Cube

Using Gauss's Law

§ 6.3
Lecture 07

Charged Rod

The rod has radius R, a volume charge density ρ. Find the electric field a distance r (r>R) from the center of the rod using Gauss's Law.

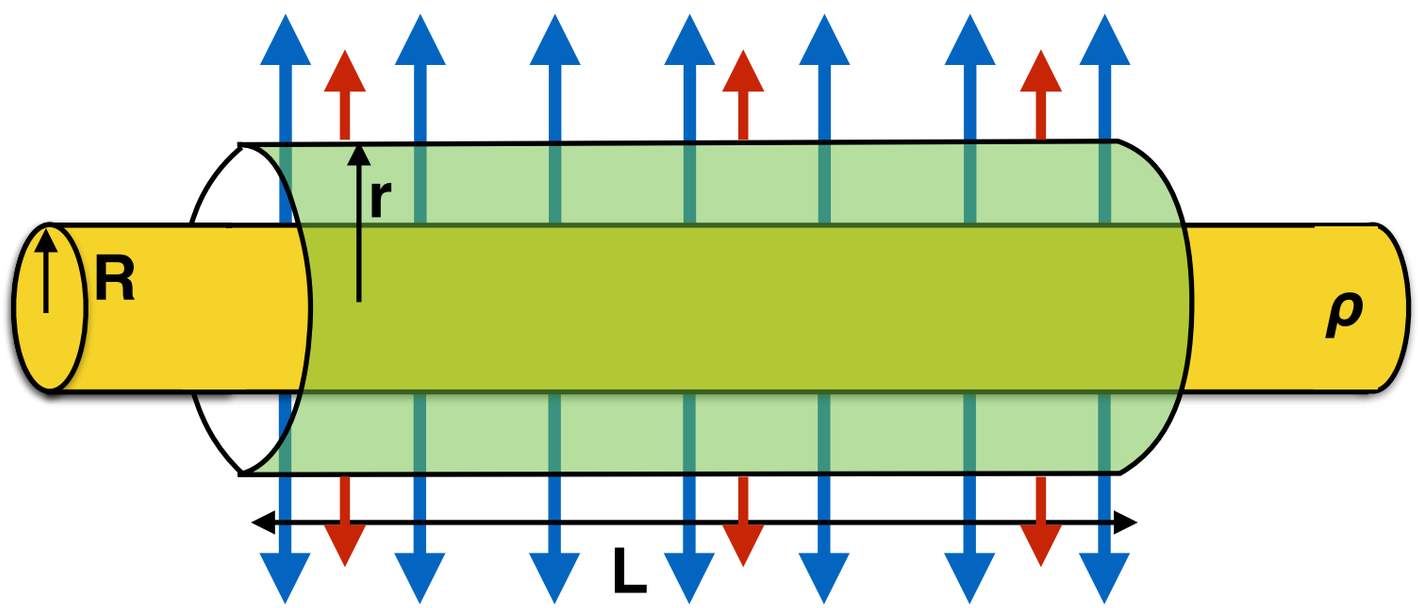


Charged Rod

Look at just the curved surface first.

$\vec{E}$ is outward from the rod

$d\vec{A}$ is outward from the surface

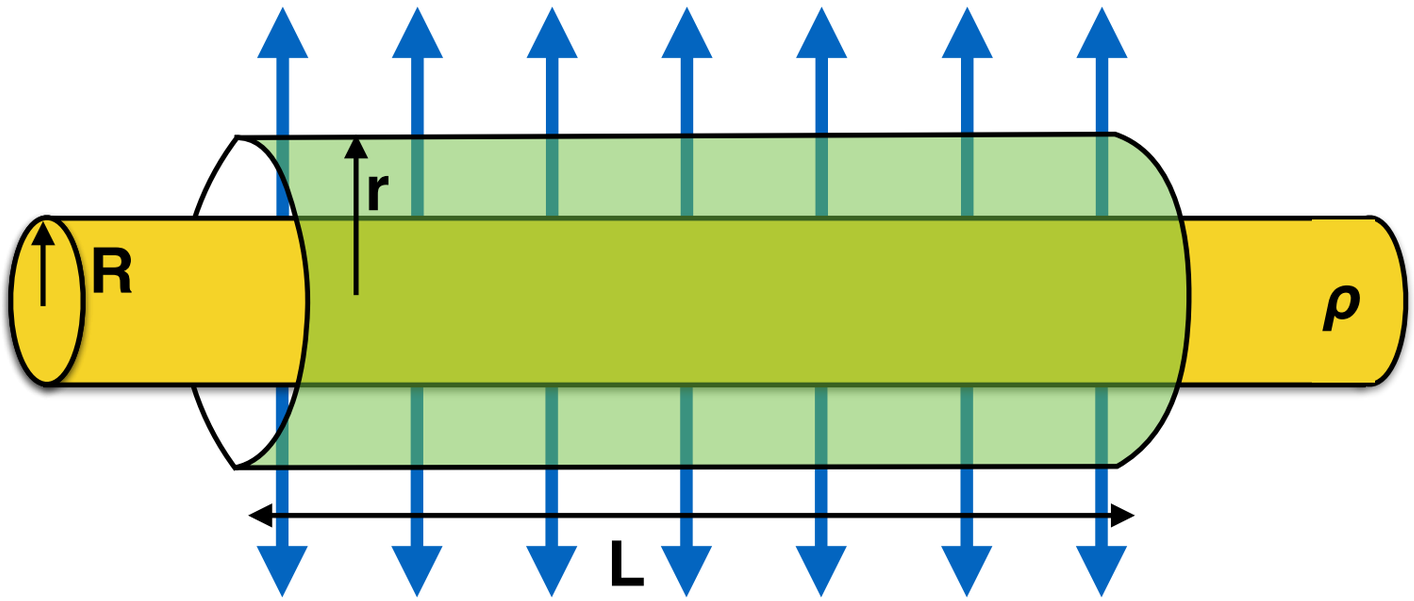


$$\int \vec{E} \cdot d\vec{A} = \int E \, dA$$

Charged Rod

Curved Surface Flux

The gaussian surface has length L and radius r. What is the flux through the <u>curved face</u> of the gaussian cylinder?



1. $\Phi_E = 0$
2. $\Phi_E = E \times 2\pi r L$
3. $\Phi_E = E \times \pi r^2$
4. $\Phi_E = E \times \pi r^2 L$
5. $\Phi_E = E \times L^2$
0. Don't wait on my response

$$\int \vec{E} \cdot d\vec{A} = \int E\, dA$$

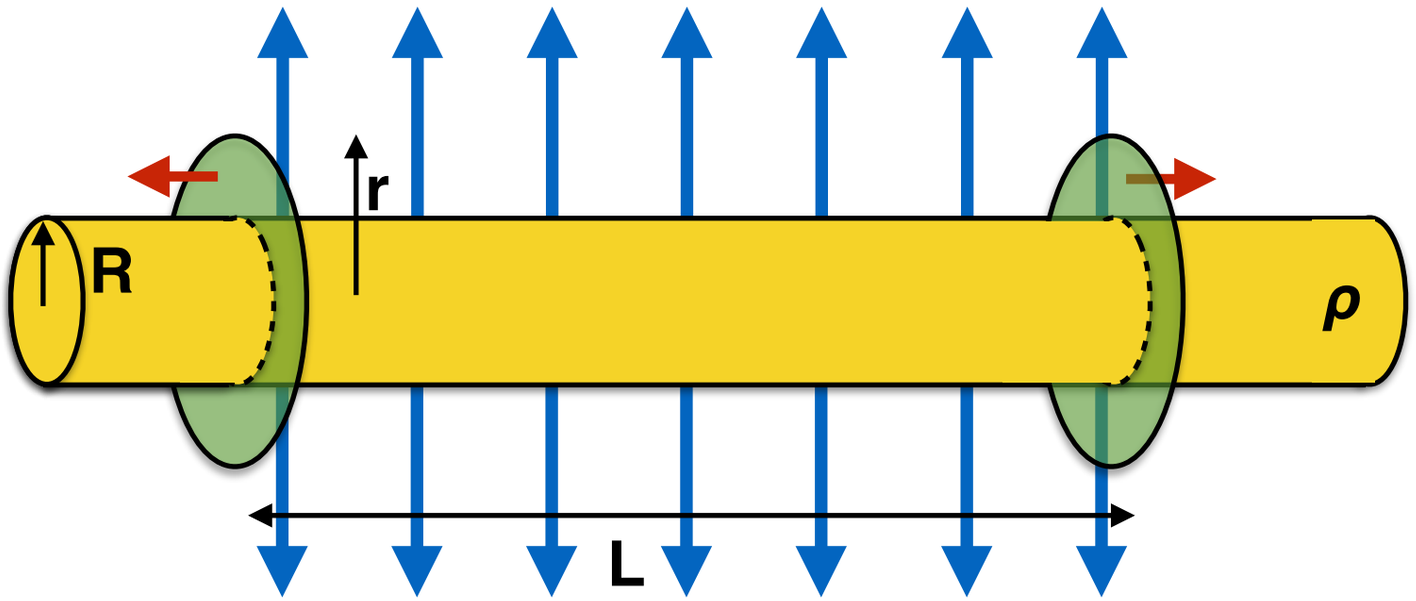
$$= E \int dA$$

$$= E\, A_{cylinder}$$

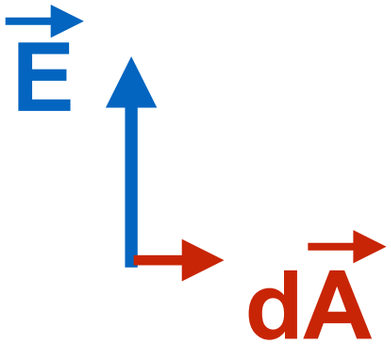
Charged Rod

Curved Surface Flux

For the two end caps, what is the direction for dA?



1. Outward from the rod
2. Inward toward the rod
3. Along the rod (right/left)
0. Don't wait on my response

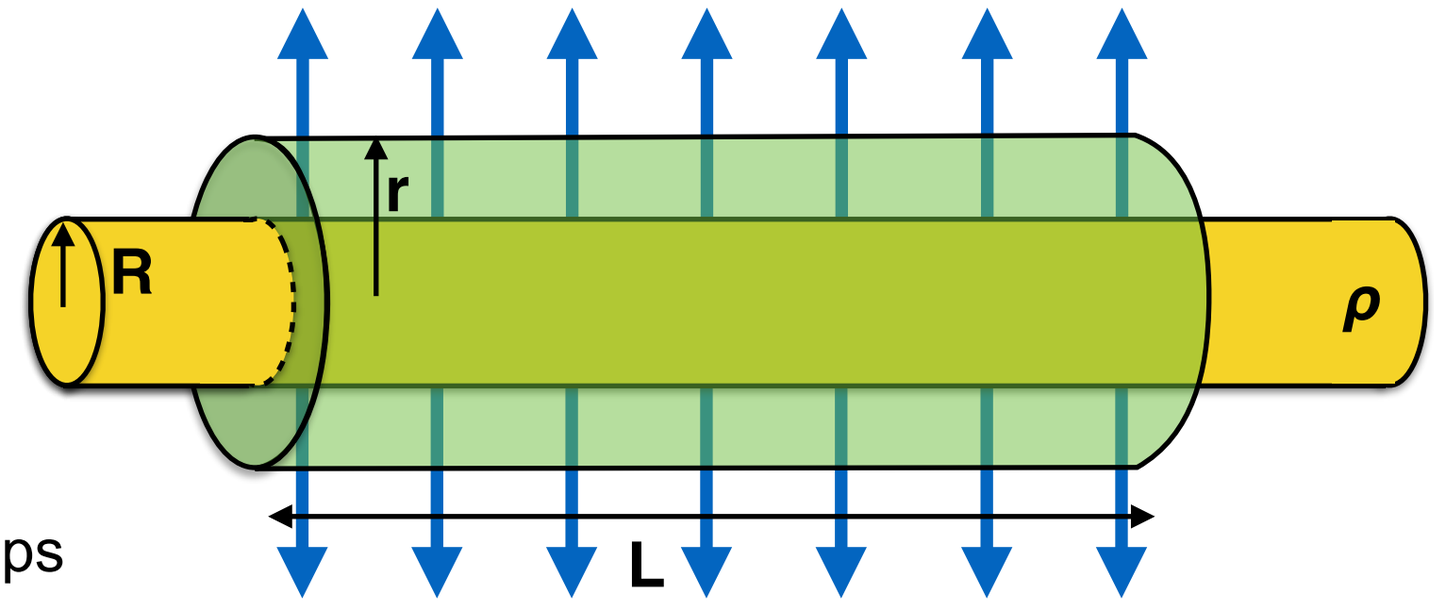


There is no flux through flat "end caps".

Charged Rod

Calculation

$$\frac{q}{\varepsilon_0} = \oint \vec{E} \cdot d\vec{A}$$



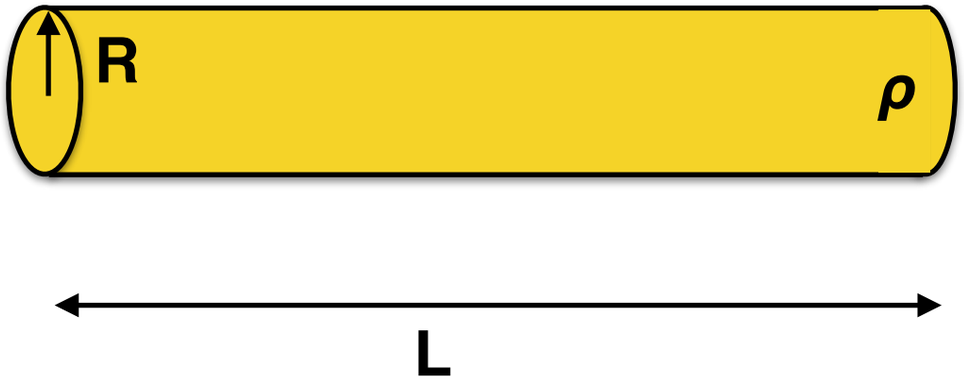
$$\oint \vec{E} \cdot d\vec{A} = \Phi_{\text{curve}} + \Phi_{\text{end caps}}$$

$$\frac{q}{\varepsilon_0} = E \times 2\,\pi\, r\, L$$

Charged Rod

Charge inside the Cylinder

$$\frac{q}{\varepsilon_0}$$



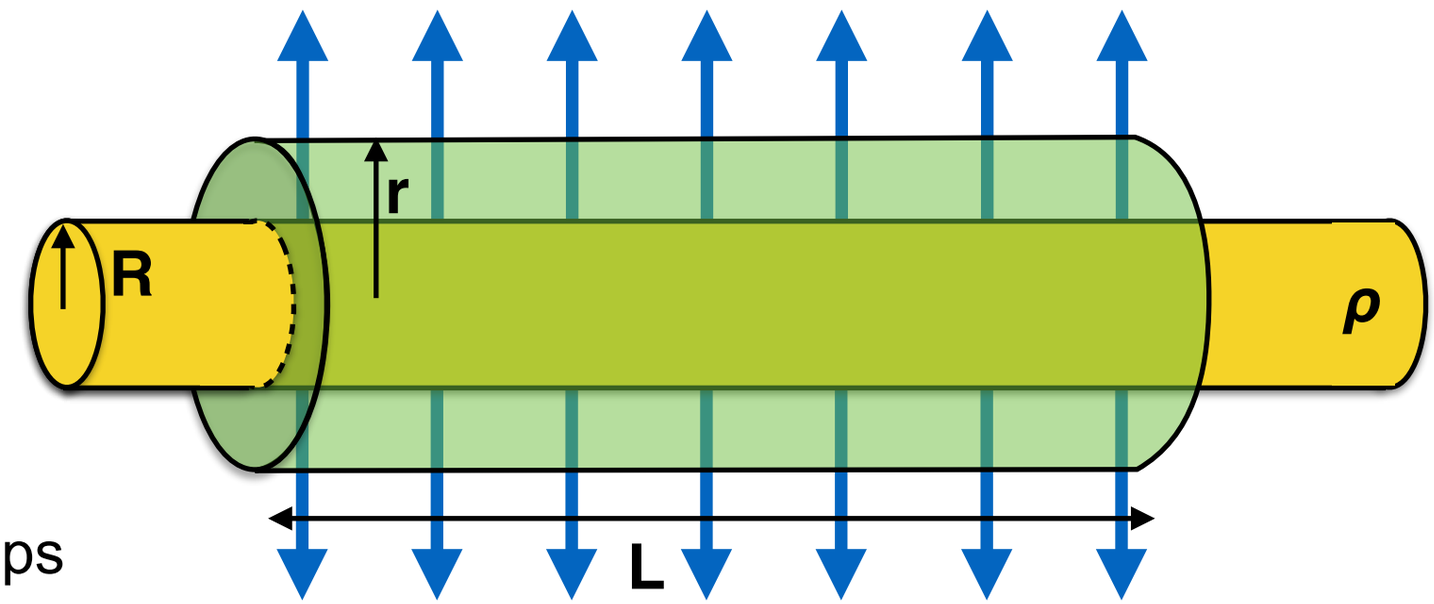
q is only the charge inside the Gaussian cylinder.

$$\rho = q / V \longrightarrow q = \rho V = \rho (\pi R^2 L)$$

Charged Rod

Final Calculation

$$\frac{q}{\varepsilon_0} = \oint \vec{E} \cdot d\vec{A}$$



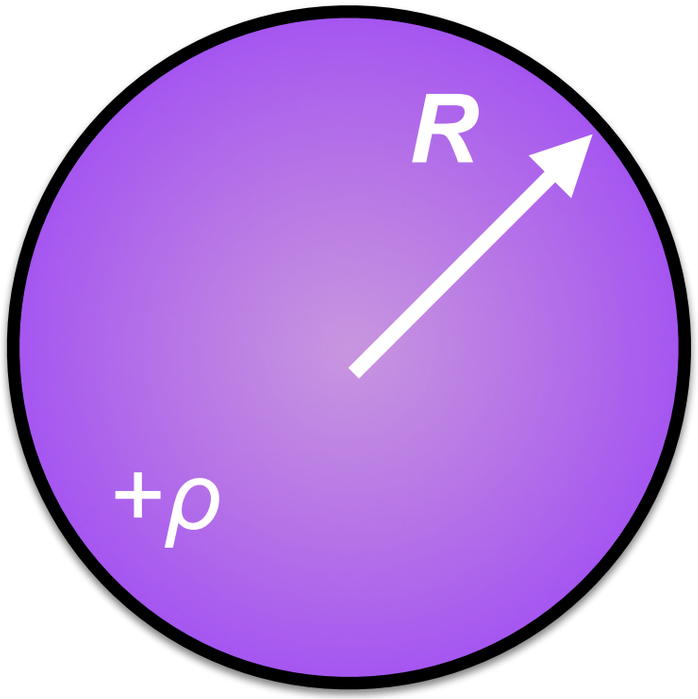
$$\oint \vec{E} \cdot d\vec{A} = \Phi_{curve} + \Phi_{end\ caps}$$

$$\frac{q}{\varepsilon_0} = E \times 2\,\pi\, r\, L$$

$$\frac{\rho\,(\cancel{\pi}\, R^2\, \cancel{L})}{\varepsilon_0} = E \times 2\,\cancel{\pi}\, r\, \cancel{L} \longrightarrow E = \frac{\rho\, R^2}{2\,\varepsilon_0\, r}$$

Insulating Sphere

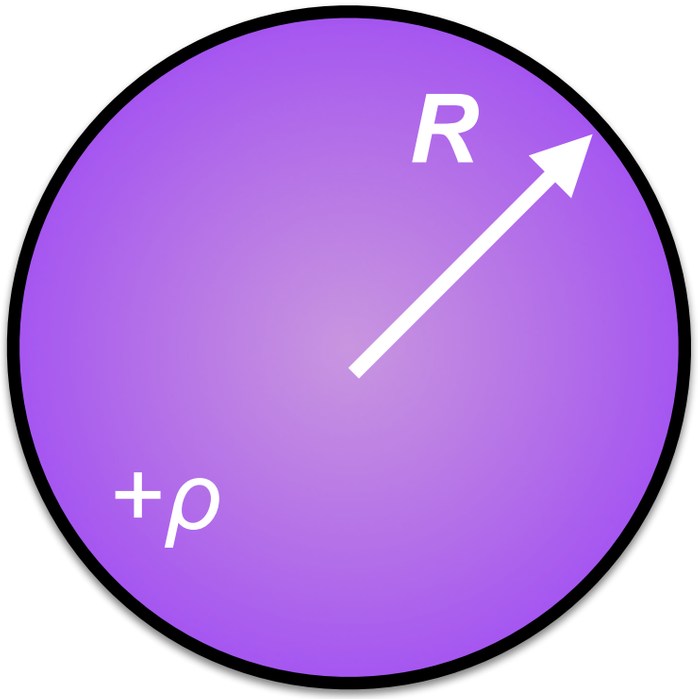
A sphere has charge density $+\rho$ and radius R? Graph E from the center of the sphere to a distance 3R from the center.



Insulating Sphere

Inside : E-direction

A sphere has charge density +ρ and radius *R?* Graph E from the center of the sphere to a distance 3R from the center.

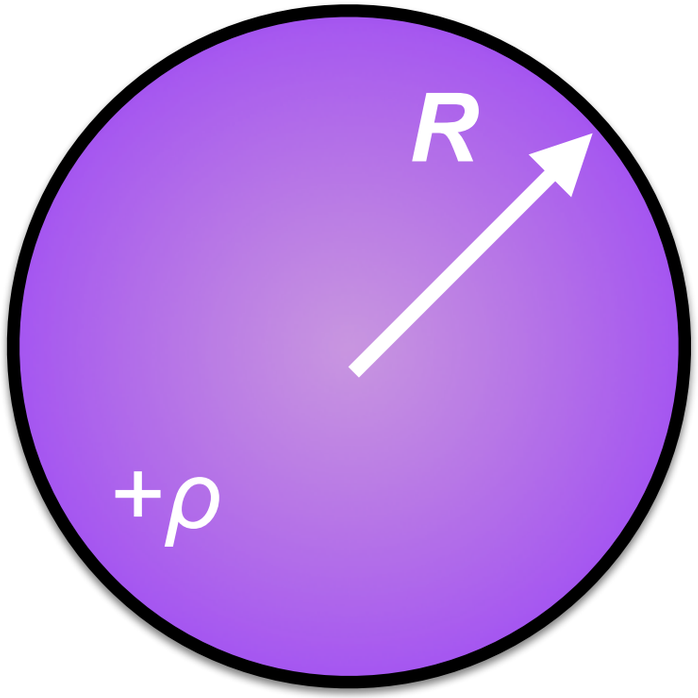


What direction do you think E will be in?

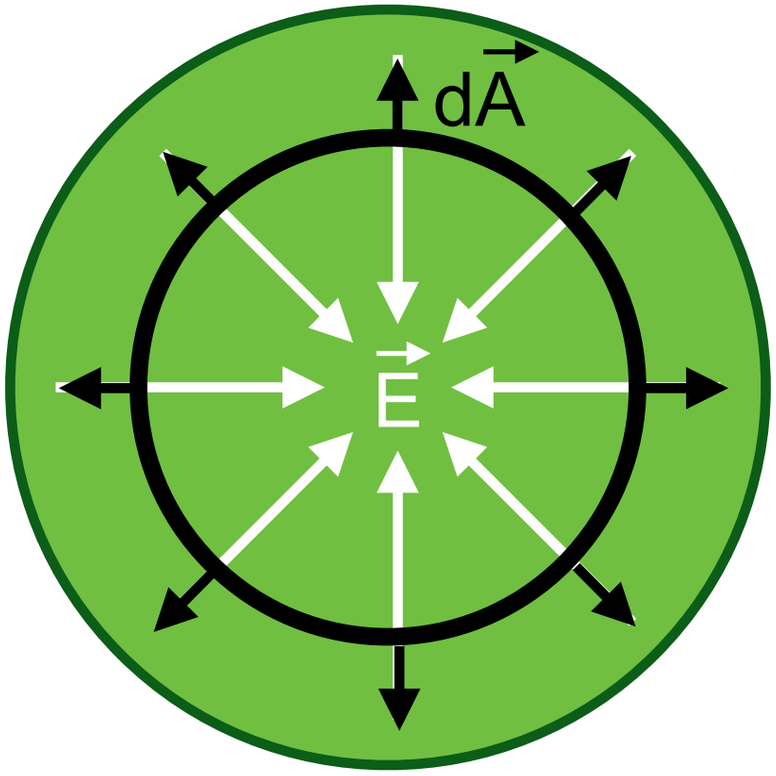
Insulating Sphere

Inside : Total Flux

A sphere has charge density $+\rho$ and radius *R?* Graph E from the center of the sphere to a distance 3R from the center.



Spherical Gaussian Surface (radius r)



$$\oint \vec{E} \cdot d\vec{A}$$

$$= \oint E \, dA \cos(180°)$$

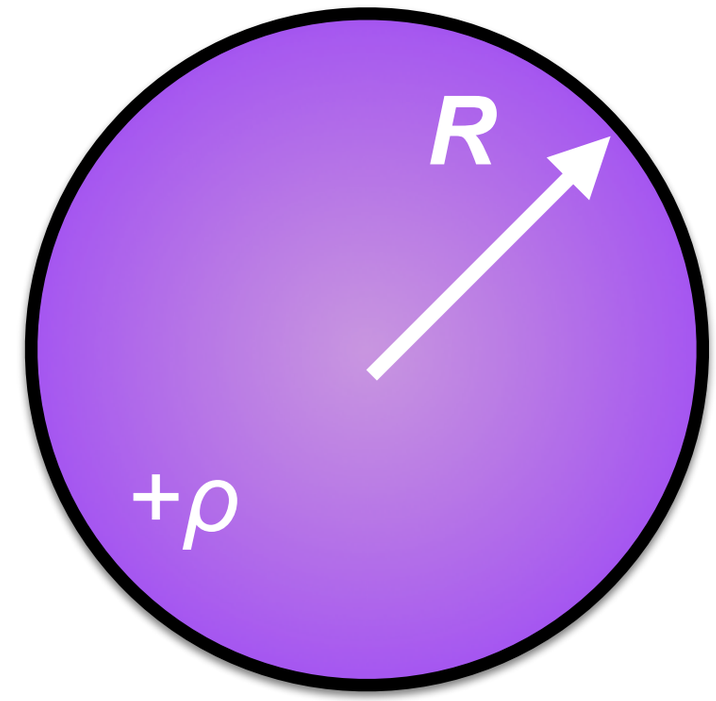
$$= -E \oint dA$$

$$= -E \, A_{\text{Gaussian sphere}} = -E \, (4 \pi r^2)$$

Insulating Sphere

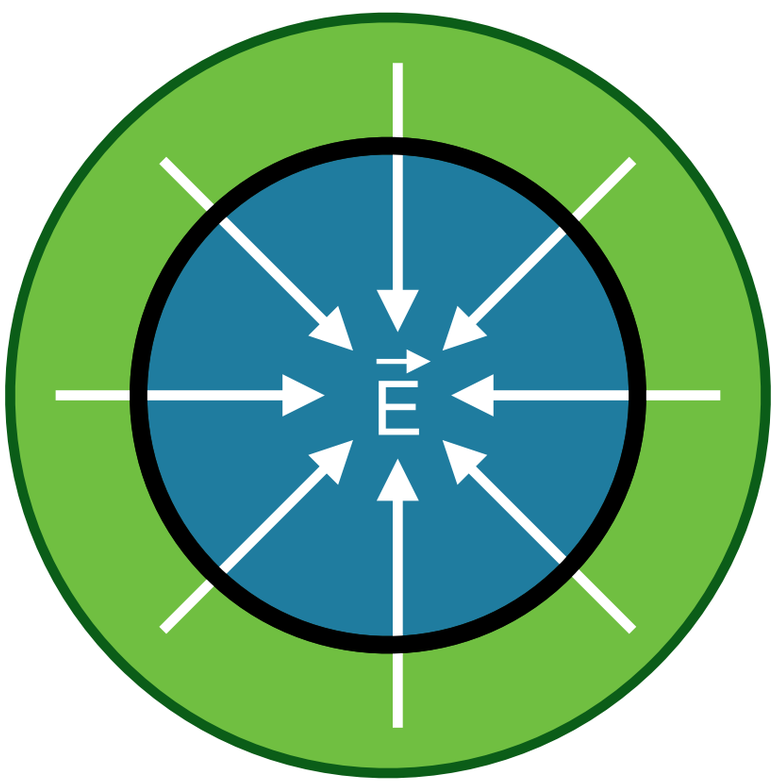
Inside : Gauss's Law

A sphere has charge density $+\rho$ and radius *R?* Graph E from the center of the sphere to a distance 3R from the center.



Use Gauss's Law!

$$\frac{q}{\varepsilon_0} = \oint \vec{E} \cdot d\vec{A}$$



$$\frac{q_{enclosed}}{\varepsilon_0} = -E\,(4\,\pi\,r^2)$$

$$\frac{\rho\, V_{Gaussian\ Sphere}}{\varepsilon_0} = -E\,(4\,\pi\,r^2)$$

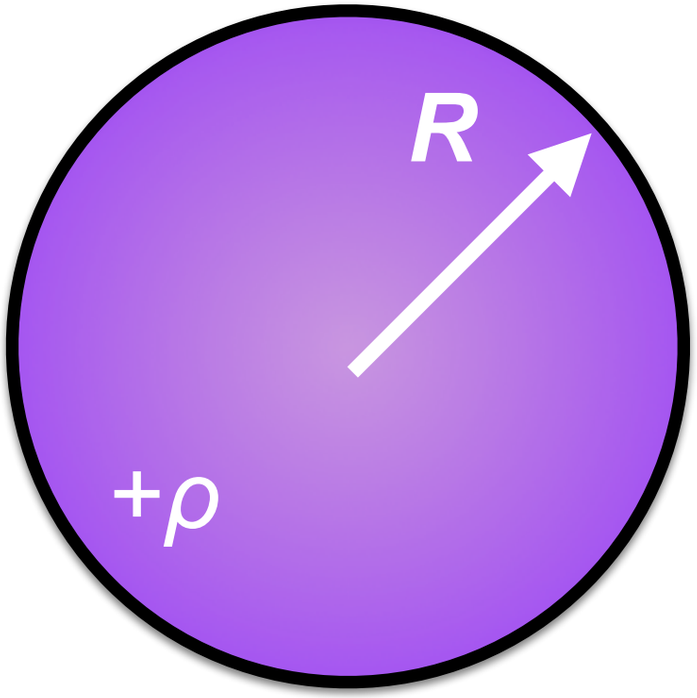
$$\frac{\rho\,(\frac{4}{3})\,\pi\,r^3}{\varepsilon_0} = -E\,(4\,\pi\,r^2) \longrightarrow E = \frac{-\rho\,r}{3\,\varepsilon_0}$$

???

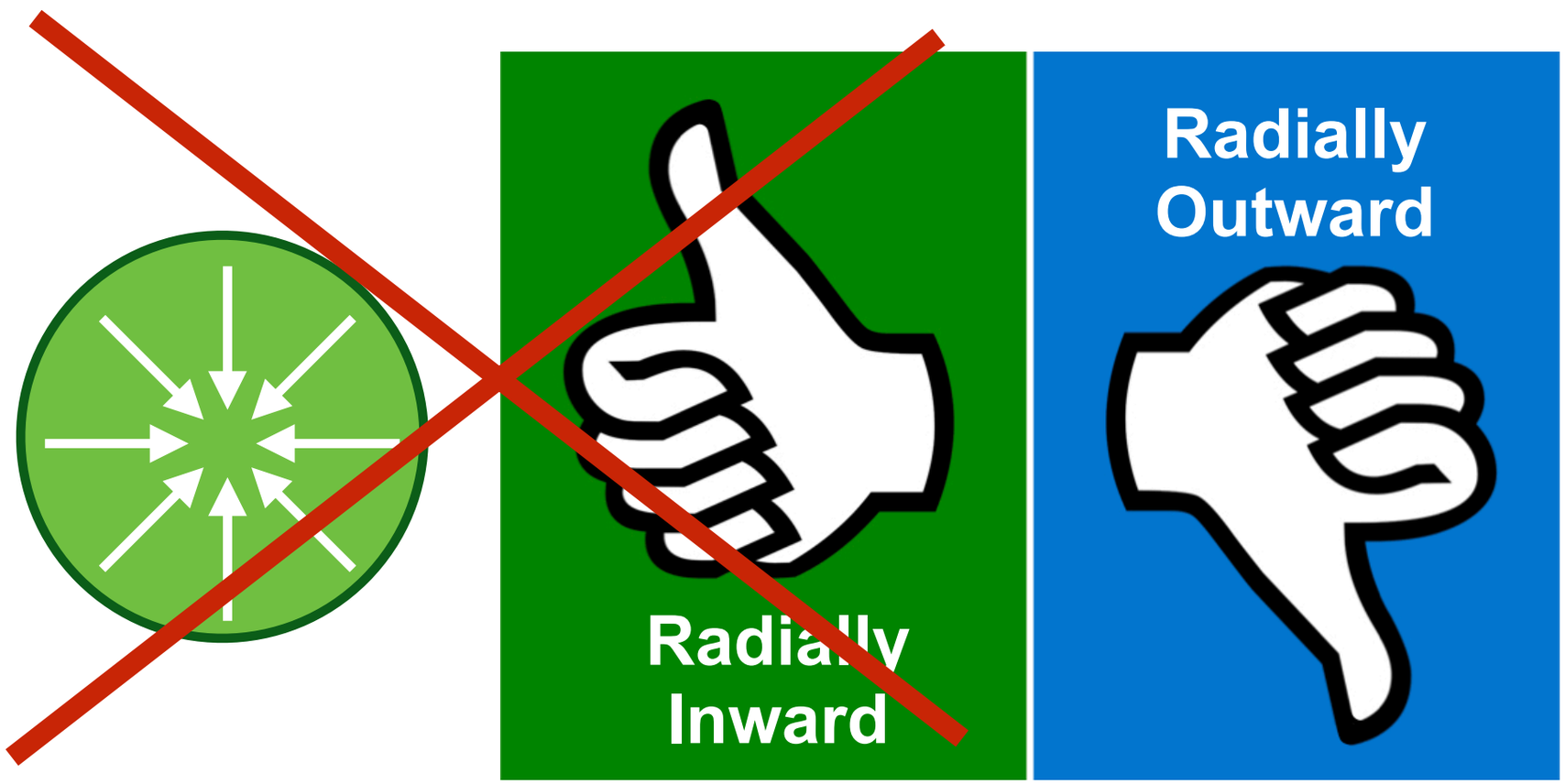
Insulating Sphere

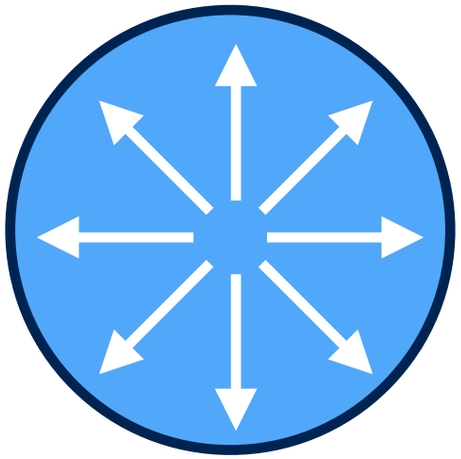
Inside : E-direction - Redo

A sphere has charge density +ρ and radius *R?* Graph E from the center of the sphere to a distance 3R from the center.



What direction do you think E will be in?

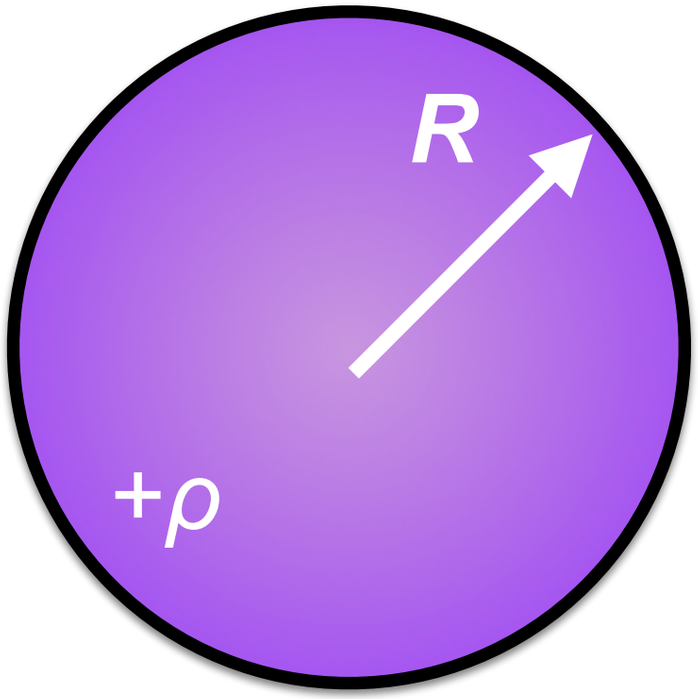


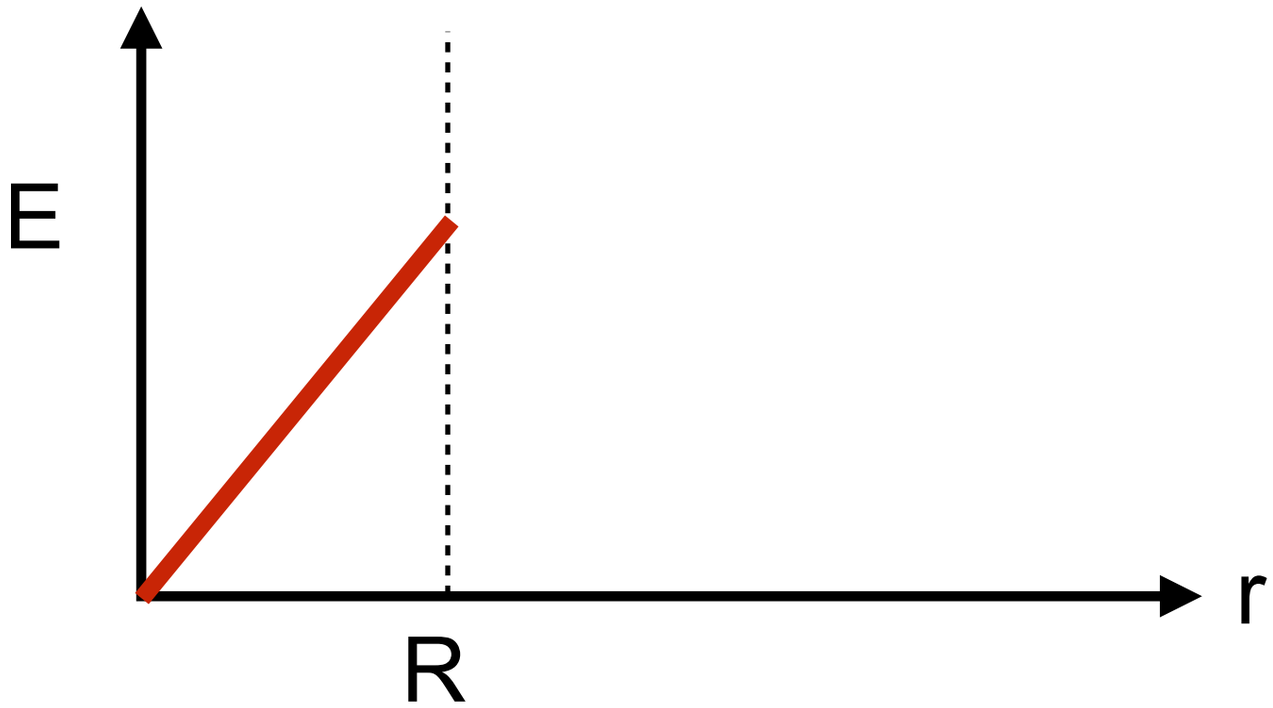


Insulating Sphere

Inside - Graph

A sphere has charge density $+\rho$ and radius *R?* Graph E from the center of the sphere to a distance 3R from the center.



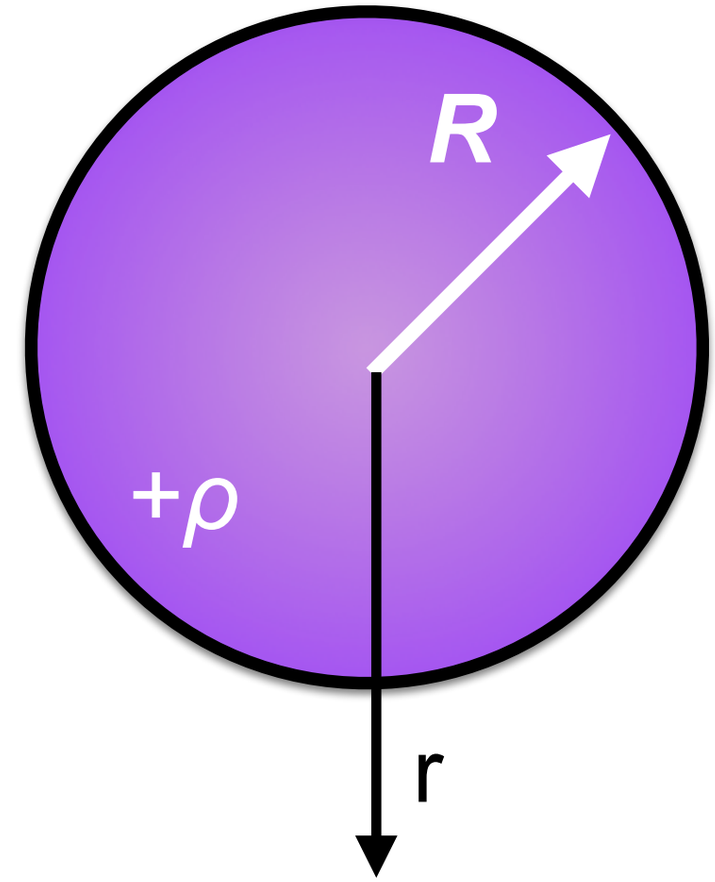


<u>Inside</u> $E = \dfrac{\rho\, r}{3\, \varepsilon_0}$

Insulating Sphere

Outside the Sphere - Equation

A sphere has charge density $+\rho$ and radius *R?* Graph E from the center of the sphere to a distance 3R from the center.



We found this in the entry video :

$$E = \frac{Q}{4\,\pi\,\varepsilon_0\,r^2}$$

What's Q?

$$Q = \rho\, V_{sphere}$$

$$Q = \rho\,\left(\tfrac{4}{3}\right)\pi\,R^3$$

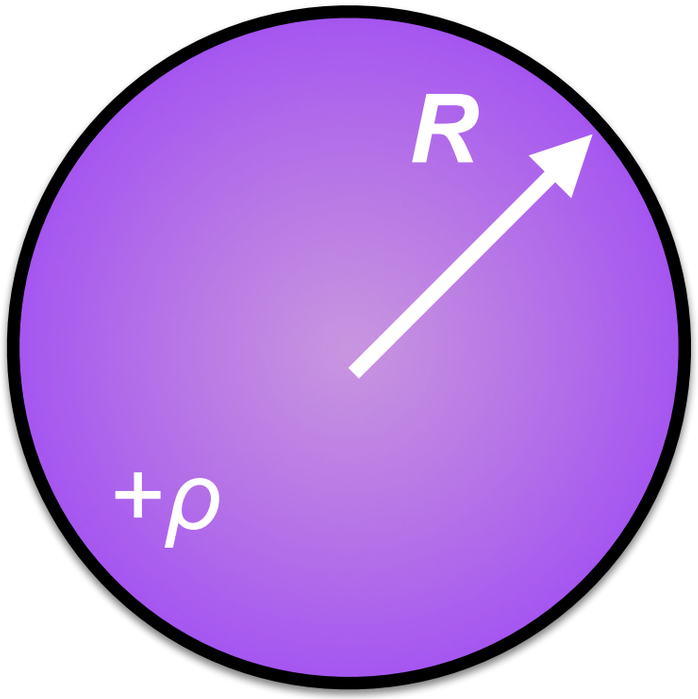
$$E = \frac{\rho\,\left(\tfrac{4}{3}\right)\pi\,R^3}{4\,\pi\,\varepsilon_0\,r^2}$$

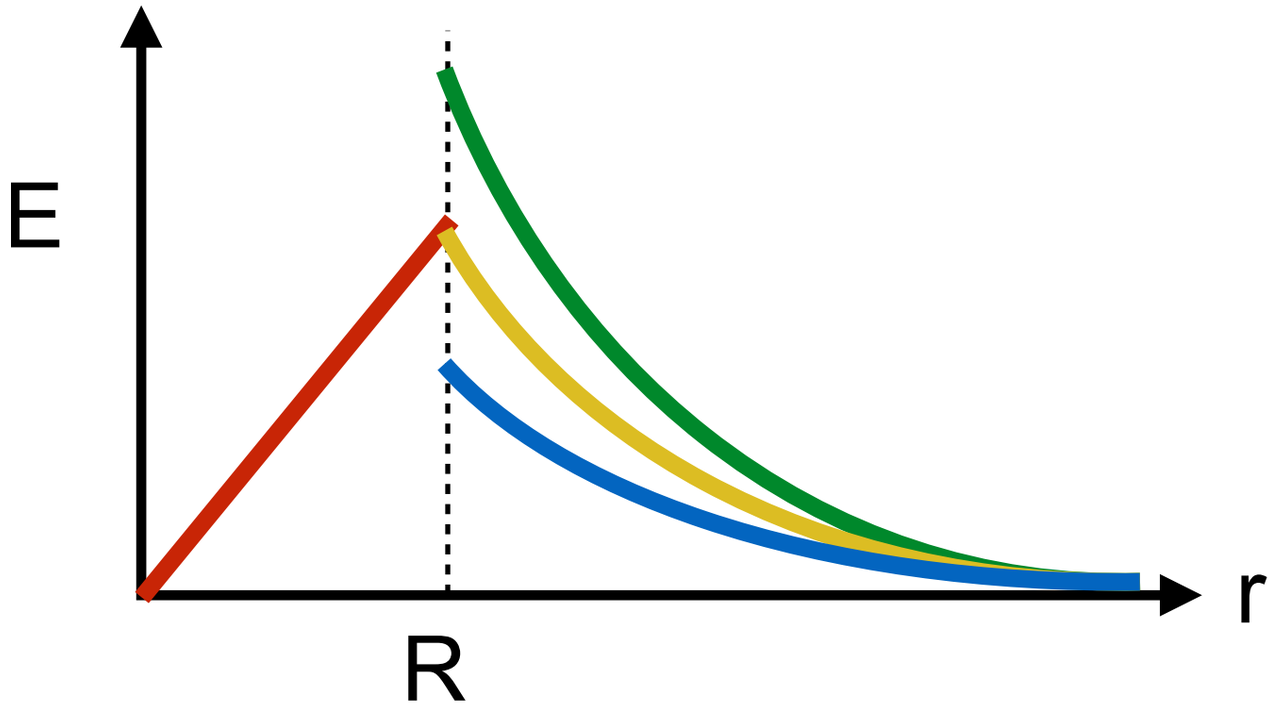
$$E = \frac{\rho\,R^3}{3\,\varepsilon_0\,r^2}$$

Insulating Sphere

Outside a Sphere - Graph

A sphere has charge density $+\rho$ and radius *R?* Graph E from the center of the sphere to a distance 3R from the center.





<u>Inside</u> $E = \dfrac{\rho\, r}{3\, \varepsilon_0}$

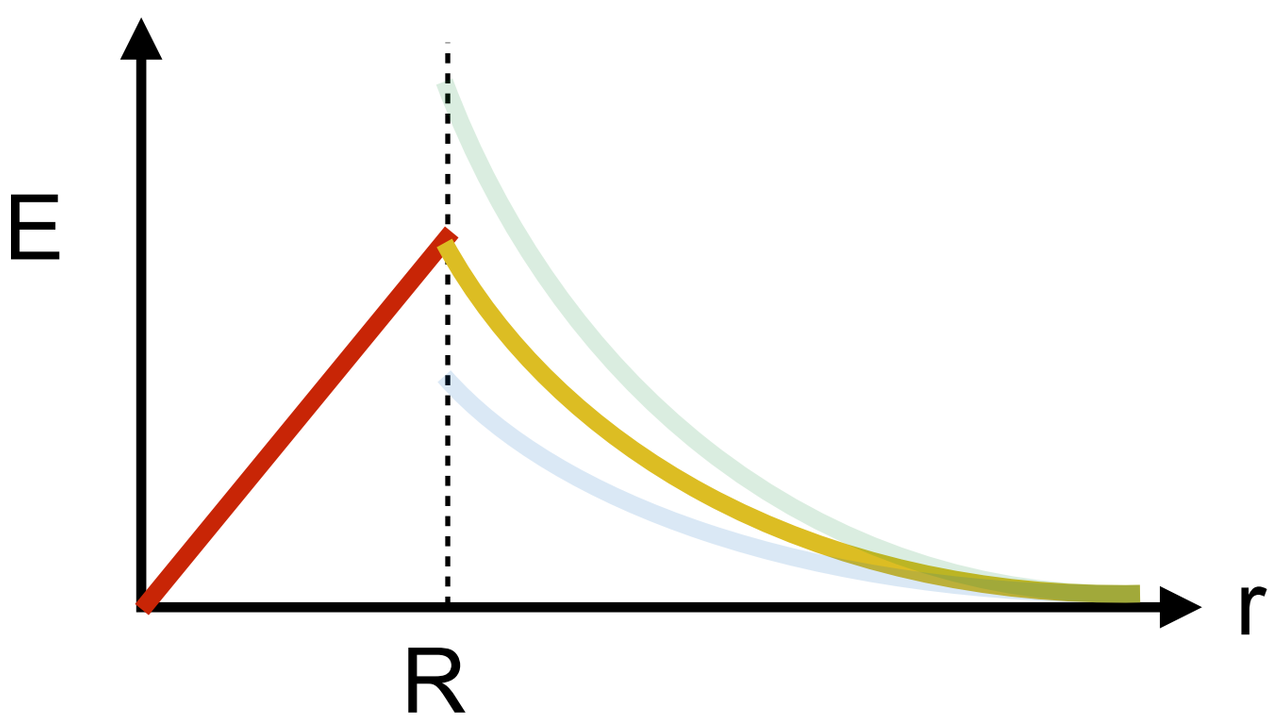
<u>Outside</u> $E = \dfrac{\rho\, R^3}{3\, \varepsilon_0\, r^2}$

Insulating Sphere

Surface of the Sphere

At r=R, which equation is larger?

1. Outside
2. They are equal
3. Inside
0. Don't wait on my response



<u>Inside</u> $E = \dfrac{\rho\, r}{3\, \varepsilon_0}$

<u>Outside</u> $E = \dfrac{\rho\, R^3}{3\, \varepsilon_0\, r^2}$

Pre-class Slides

Gauss's Law

§ 6.2
Lecture 07

Youtube Playlist

Why Gauss's Law?

A method to calculate electric field that

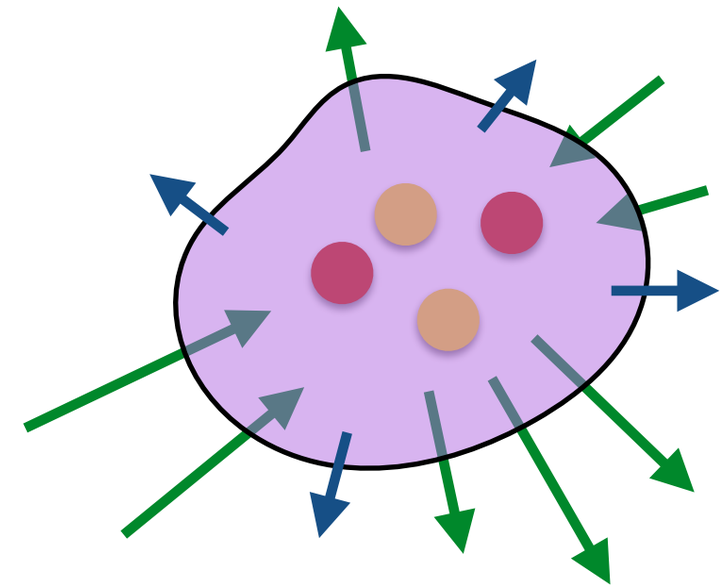
is mathematically elegant,

makes powerful use of symmetry, and

helps us visualize electric fields.

Gauss's Law Equation

Charge : **q**



Closed surface surrounding the charge (**Gaussian surface**)

Total flux passing through the Gaussian surface:

$$\oint \vec{E} \cdot d\vec{A}$$

Field passing through surface

Integral over entire, closed surface

Area vector pointing away from surface

Gauss's Law : $$\frac{q}{\varepsilon_0} = \oint \vec{E} \cdot d\vec{A}$$

Permittivity of Free Space

$$\varepsilon_0 = 8.85\times10^{-12}\ C^2/Nm^2$$

Related to Electrostatic Constant : $K = 1 / (4\pi\varepsilon_0)$

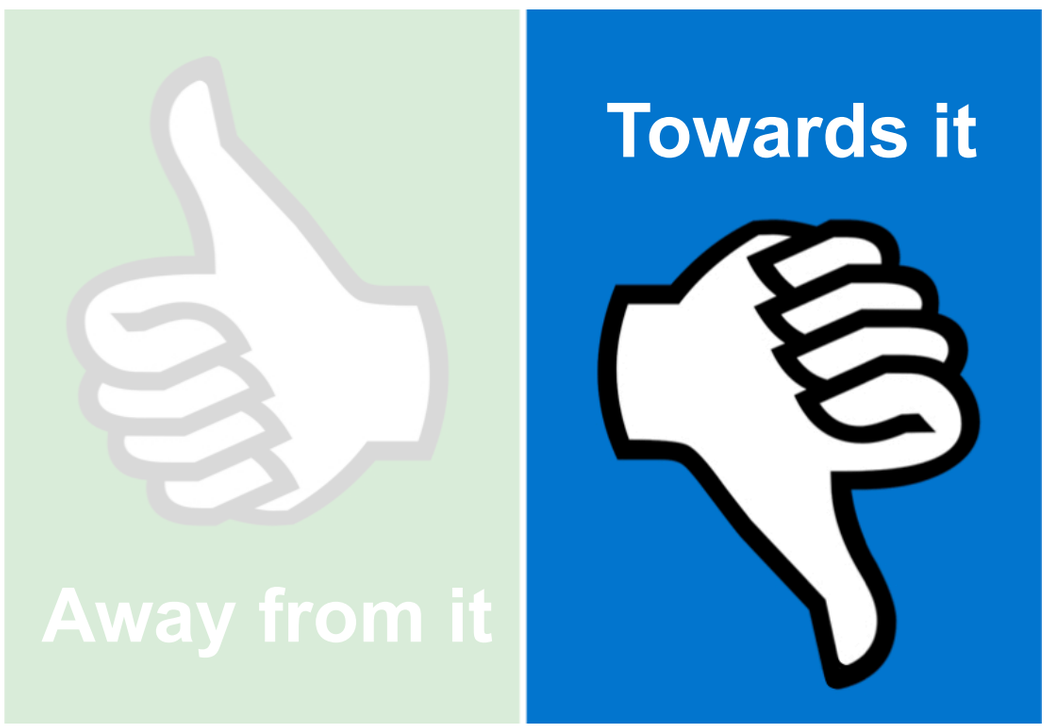
$$F = K\frac{|q_1|\ |q_2|}{r^2} = \frac{1}{4\pi\varepsilon_0}\frac{|q_1|\ |q_2|}{r^2}$$

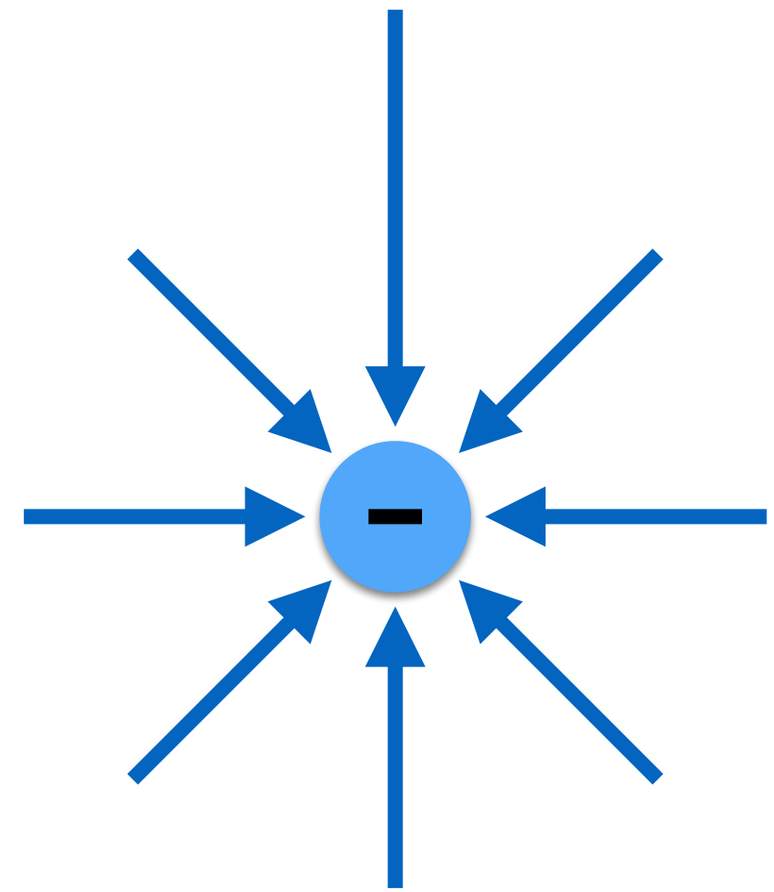
Permittivity of free space is more fundamental.
Electrostatic constant is easier to work with.

Knowledge Recall

Negative Charges

A negative charge creates an electric field that point…





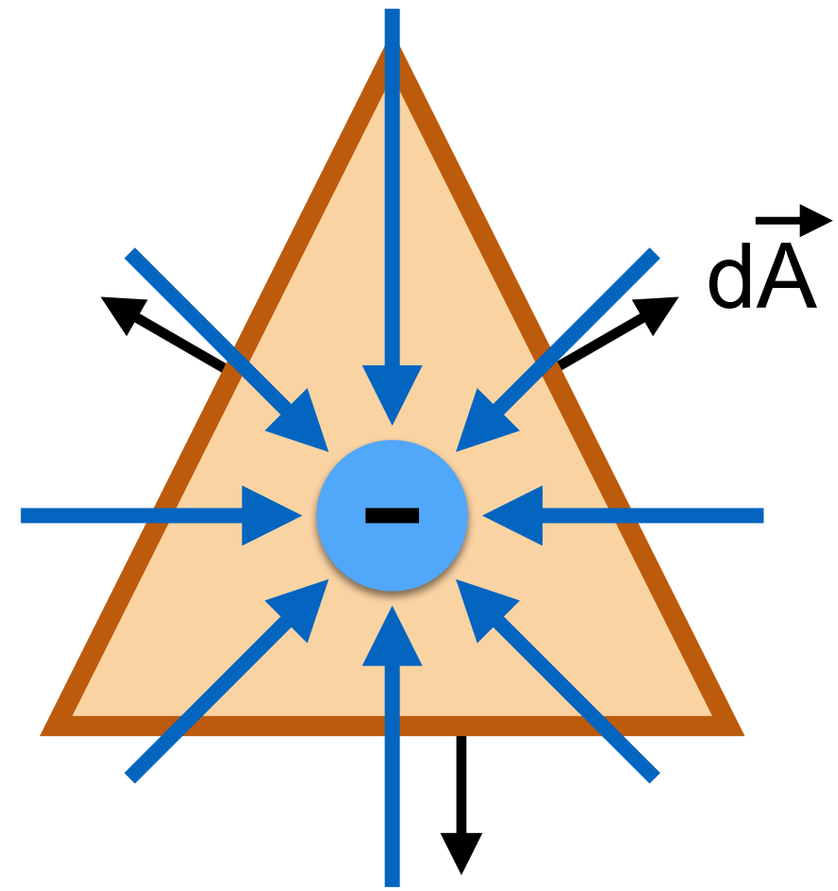
0. Don’t wait on my response

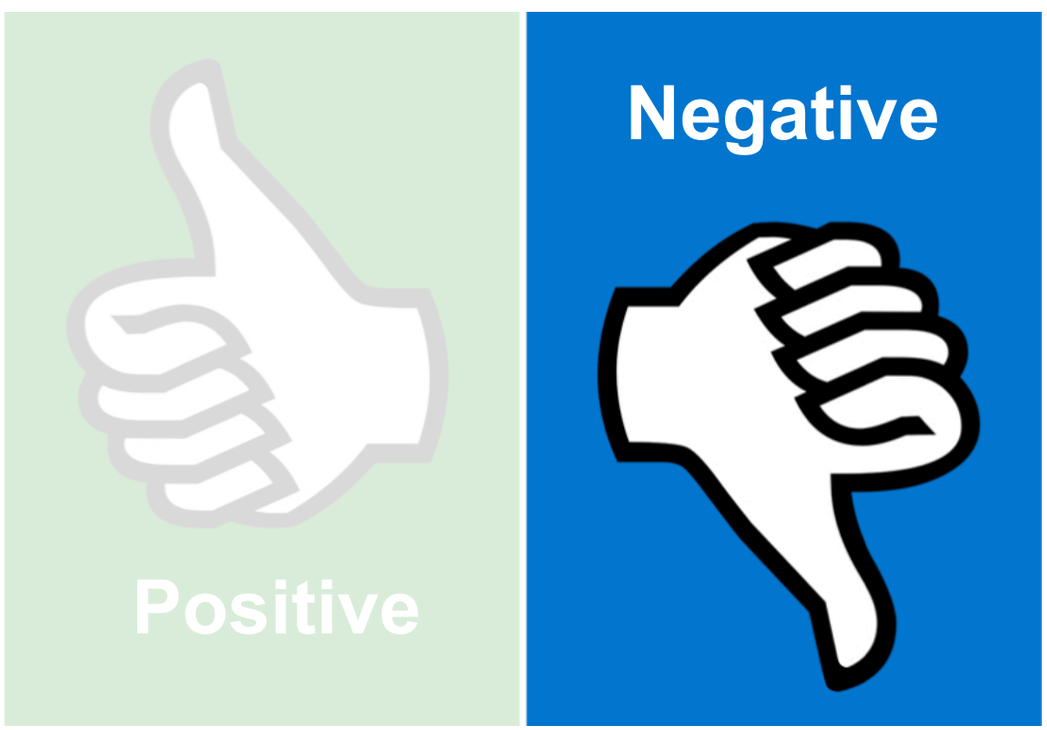
Flux & Gauss's Law

Charge and Sign

$d\vec{A}$ always points outward from a Gaussian surface.

If electric field points into a surface, then $\vec{E} \cdot d\vec{A}$ is…





0. Don't wait on my response

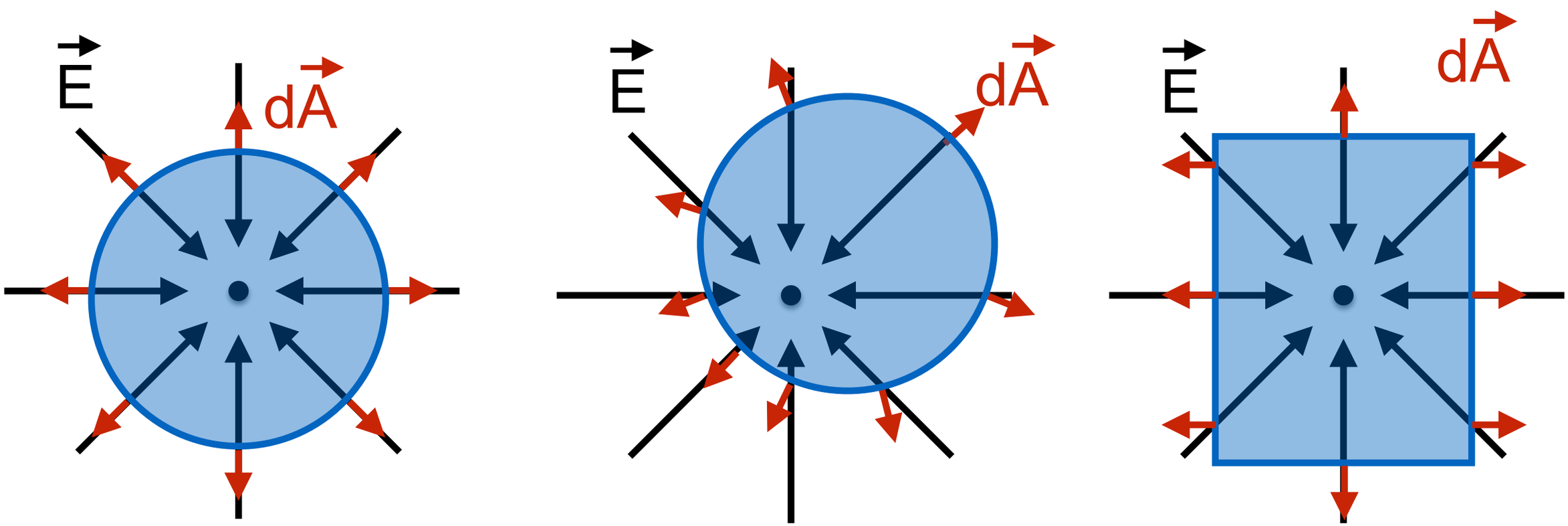
Field *entering* the surface gives a negative flux.

Flux & Gauss's Law

For Gauss's Law, the flux is calculated for a closed surface surrounding charge.

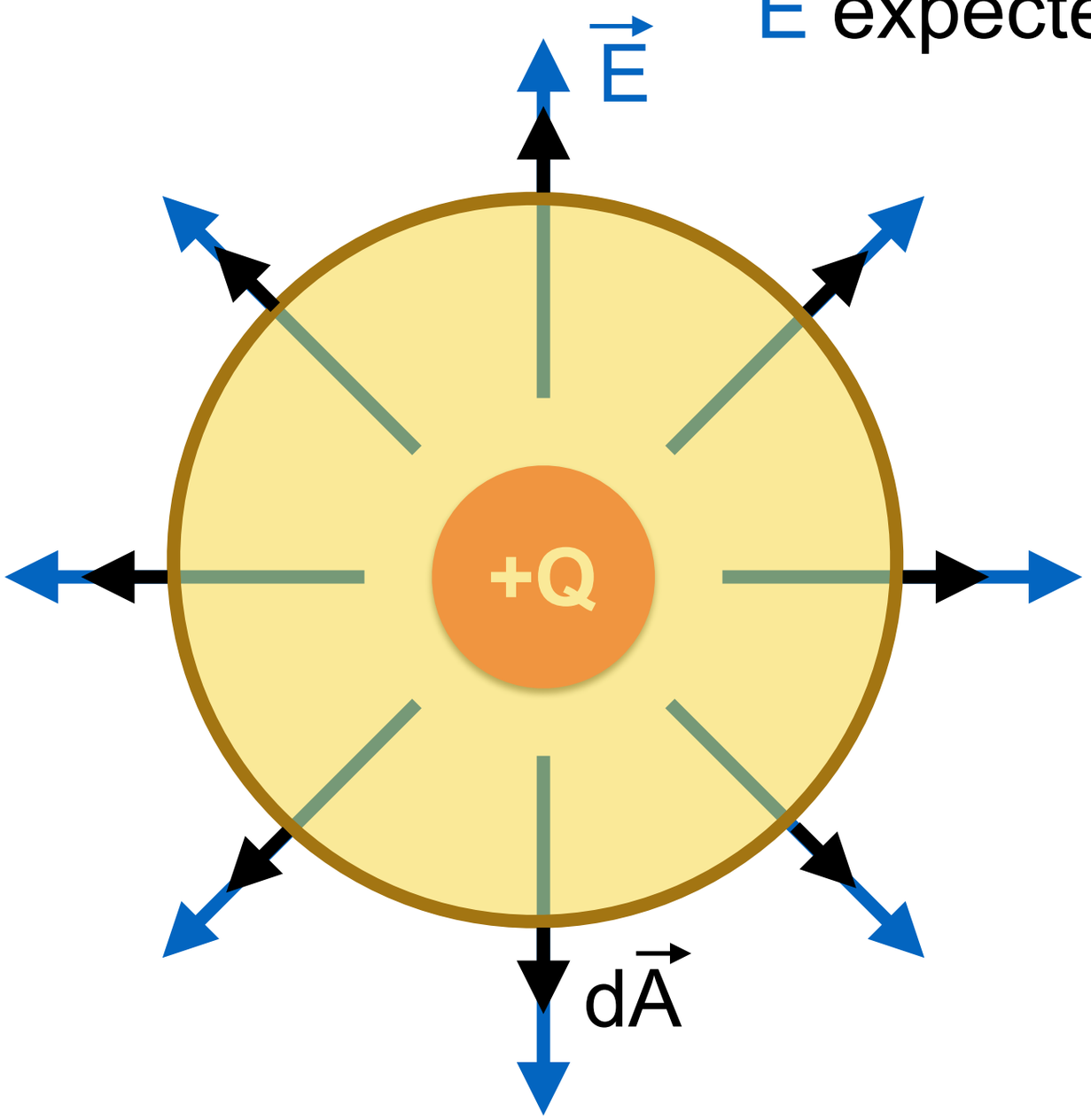
$$\frac{q}{\varepsilon_0} = \oint \vec{E} \cdot d\vec{A}$$

The closed surface is called a **Gaussian** Surface. It can be any shape, but some are better than others.



It's important to pick a *clever* surface to use Gauss's Law.

Outside a Charged Sphere



E expected to: Be radially outward

Have same magnitude at equal distances from the sphere

$$\frac{q}{\varepsilon_0} = \oint \vec{E} \cdot d\vec{A}$$

"Nice" Surface?

Want $\vec{E}$ to be parallel, anti-parallel, or perpendicular to $d\vec{A}$ everywhere on surface.

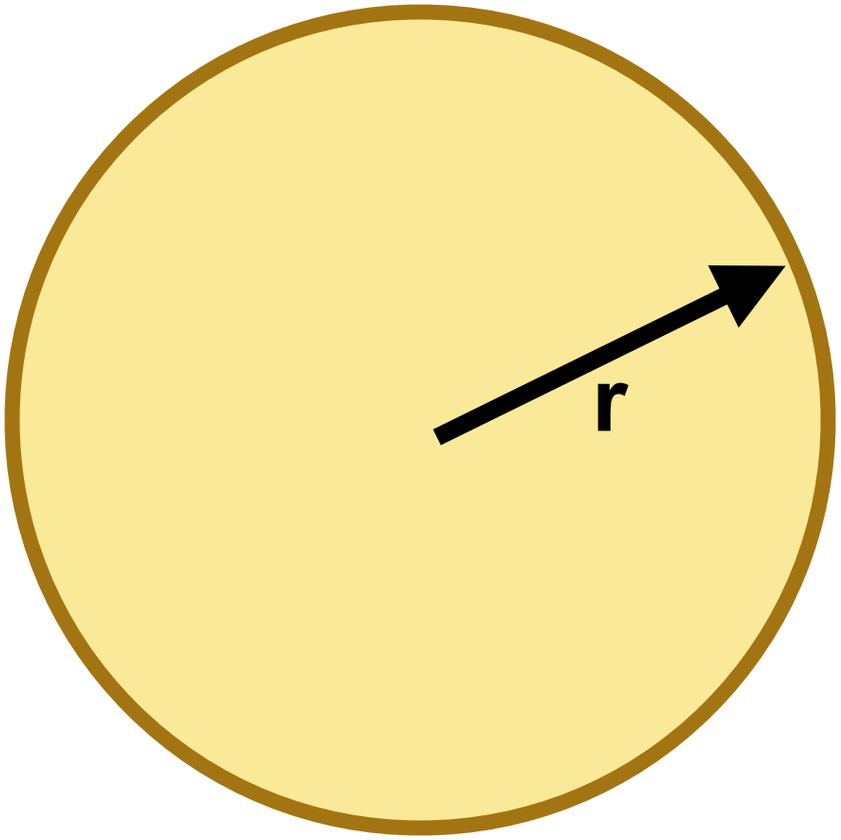
$$\oint \vec{E} \cdot d\vec{A} = \oint E\, dA = E \oint dA = E\, A_{sphere}$$

Outside a Charged Sphere

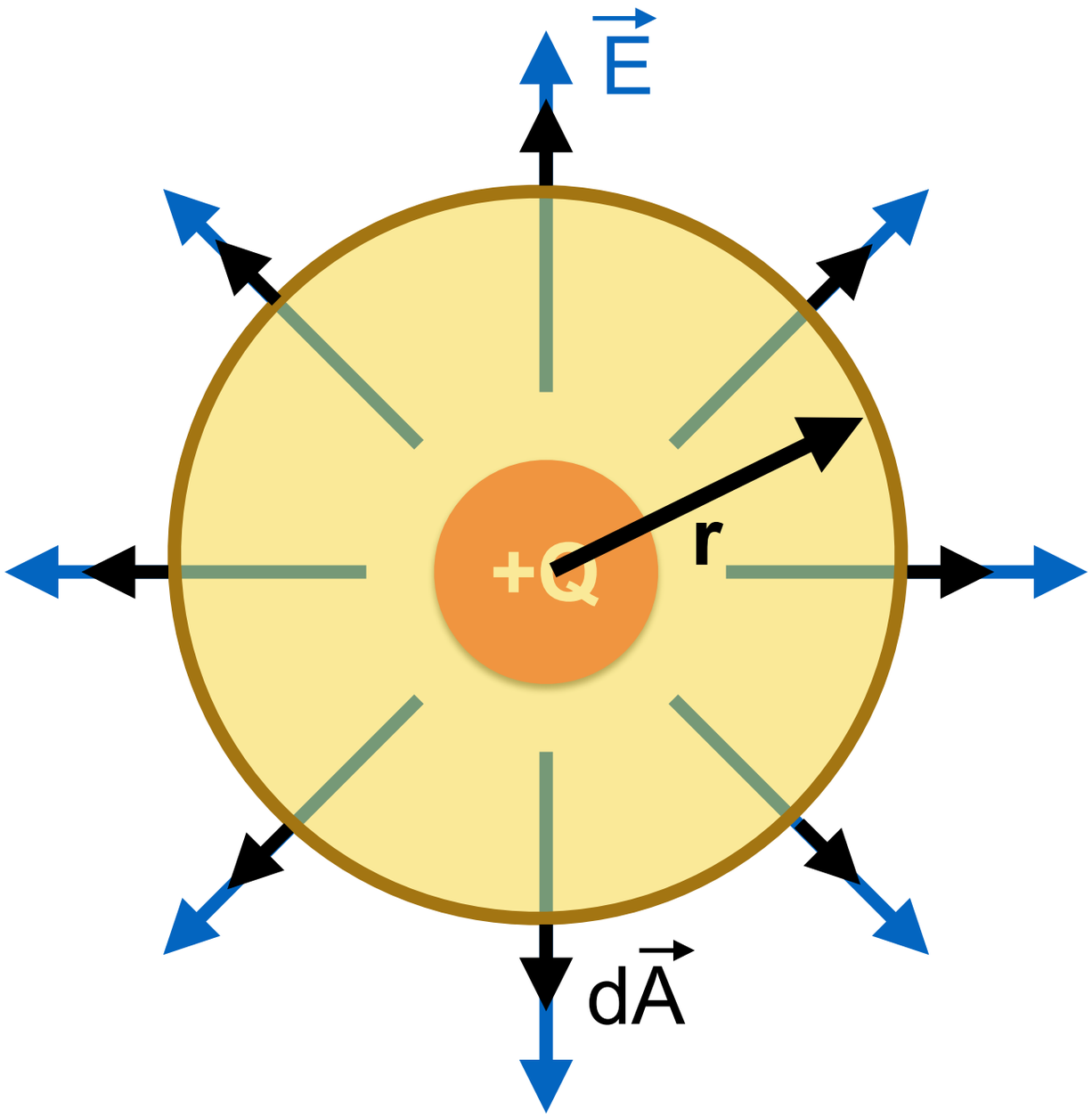
Surface Area

The surface area of a sphere with radius r is…

1. πr^2
2. $2 \pi r$
3. $2 \pi r^2$
4. $4 \pi r^2$
5. $(4/3) \pi r^3$
0. Don't wait on my response



Outside a Charged Sphere



$$\frac{q}{\varepsilon_0} = \oint \vec{E} \cdot d\vec{A}$$

$$\frac{Q}{\varepsilon_0} = E\, A_{sphere}$$

$$\frac{Q}{\varepsilon_0} = E\,(4\,\pi\, r^2)$$

$$E = \frac{Q}{\varepsilon_0\,(4\,\pi\, r^2)}$$

Magnitude

Direction : Radially away from sphere

Using Gauss's Law

To solve for **E** directly, one **must**

1. be able to determine from the symmetry of the charge distribution what direction **E** points and on what variables **E** depends so that one can,
2. create a Gaussian surface on which **E·dA** is a constant. Once such a Gaussian surface has been created, one can then
3. solve for E by pulling it out of the integral.

Rachel Pepper, et. al. PERC Proceedings 2010